A Quick Rundown of Options

What is an option?

An option is a contract between a buyer and a seller. It gives the buyer (the owner of holder of the option) the opportunity to buy or sell the underlying asset at a specific strike price prior to or on a specific date. Options can provide investors with more opportunities than traditional equity buy/sell strategies.

Advantages and risks of options

Options are cost efficient with great leveraging power. An investor can obtain an option position similar to stock position without actually trading the stock themselves but with more volatility risk. Options is a great way to hedge your positions and have the potential to deliver percentage returns when properly understood and losses when not understood.

Types of Options

There are two types of options, a call option and a put option. Understanding what each of these are and how they work will help you determine when and how to use them. The buyer of an option pays a premium (payment) to the seller of an option for the right, not the obligation, to take delivery of the underlying futures contract (exercise). This financial value is treated as an asset, although eroding, to the option buyer and a liability to the seller.

For simplification:

- Call options allow the holder to buy the asset at a specified price (strike price) within a specific timeframe (Expiration date).
- Put options allow the holder to sell the asset at a specified price (strike price) within a specific timeframe (Expiration date).

Option contracts are composed of two key aspects: "**Strike price**" and "**Expiration date**"

1. **Strike Price:** the strike price is the price at which the buyer of a call option has the right to purchase the futures contract, or the buyer of a put option has the right to sell a futures contract. This is also referred to as the exercise price.
2. **Expiration date:** each option contract will have a specific expiration date by which the holder must exercise their option.

CALL OPTIONS	PUT OPTIONS
Call Options gives the buyer the right to **BUY** the stock at a predetermined price & date.	**Put Options** gives the buyer the right to **SELL** the stock at a predetermined price and date.
Buyers expect the stock price to **RISE**	Buyer expect the stock price to **FALL**
Max profit is **INFINITE** as the rise in stock price can't be capped. Risk is limited to the premium paid.	Max profit is **LIMITED** as fall in stock price stops at zero. Risk is limited to the premium paid.

Options Pricing

Let's talk about pricing or valuation of an option contract. The terms in this equation may seem foreign at first, but don't worry- I will break them down for you.

Option Price = Intrinsic Value + Extrinsic Value

The intrinsic value of an option is the amount that the market price is higher than the strike price for a call and lower than the strike for a put. In other words, the intrinsic value is the amount of money that the option would be worth if it expired today. For the option to have intrinsic value, the option must be in-the-money

Now... what does in-the-money mean?

- In-the-money—The futures price is above the strike price. (ITM)
- At-the-money—The futures price is at the strike price. (ATM)
- Out-of-the-money—The futures price is below the strike price. (OTM)

Top 3 factors affecting options prices:

1. Intrinsic Value – the underlying equity price in relation to the strike price
2. Time Value (Theta) – the length of time until the option expires
3. Implied Volatility Value – Expected future price fluctuations – Derived from charting

Intrinsic Value

The intrinsic value of an option can easily be calculated, but the extrinsic value of an option is impossible to estimate at any given time in the future other than expiration. This is because the extrinsic value is made of a combination of time, volatility, and demand.

Intrinsic Value = (Current Market Price - Strike Price) x 100.
The x100 is because each option contract is composed of 100 shares.

Extrinsic Value

Extrinsic value is based on a combination of the strike price, time, volatility, and demand. We like to think of extrinsic value as the “icing on the cake.” Due to the nature of its components, it is impossible to estimate extrinsic value. Beginning traders often ask questions such as, “If I buy a call and the market goes up x number of points, what will it be worth?” Unfortunately, the answer depends on factors that can’t necessarily be measured quantitatively

The extrinsic value of an option is based on a combination of the following factors:

- **Time -** The longer the amount of time until an option’s expiration, the greater the time value of a particular option will be.
- **Volatility -** If the price of the underlying futures contract is fluctuating considerably, there is both a greater profit and a greater loss potential.
- **Demand -** If the number of traders willing to buy an option at a given price is greater than the number of traders willing to sell the same option, the value of that option appreciates.

Option Greeks

Option Greeks are used to measure the risk of an option and to gauge an option’s sensitivity to the variables that make up that risk — The variables are that matter are represented by the Greek letters **Delta, Gamma, Theta, Vega**.

Basically, option Greeks are just a mathematical aggregation of those big three things we learned above- Time, Volatility, and Demand.

- **Delta:** Represents the expected change in the option value for each $ 1 change in the price of the underlying stock.
- **Gamma:** Represents the expected change in delta for each $ 1 change in the price of the underlying stock.
- **Theta:** Represents the option’s expected daily decline due to time.
- **Vega:** Represents the expected change in the option value due to changes in volatility expectations for the underlying stock.

Delta

Delta is how much the value of an option should change when the price of the underlying rises by one dollar. Delta can be a very useful to calculate probabilities for instance a deep in the money option that is virtually certain of expiring in the money (say 90%) will have a 0.90 delta. A far OTM option with virtually no chance of expiring in the money will have a 0 delta. An ATM option has about 50% probability of being in the money because there is a 50-50 chance the stock will go up or go down. Therefore, it has about a 0.50 delta. Namely, there is a linear relationship between delta and probability.

CALL OPTIONS	**PUT OPTIONS**
Call options have delta **between 0 & 1**	**Put** options have delta **between -1 & 0**
ITM Call options have a delta **between 0.50 & 1** **ATM** Call options have deltas **around 0.50** **OTM** Call options have a delta **between 0 & 0.50**	**ITM** Put options have a delta b**etween -1 & -0.50** **ATM** Put options have deltas **around -0.50** **OTM** Put options have a delta **between -0.50 & 0**

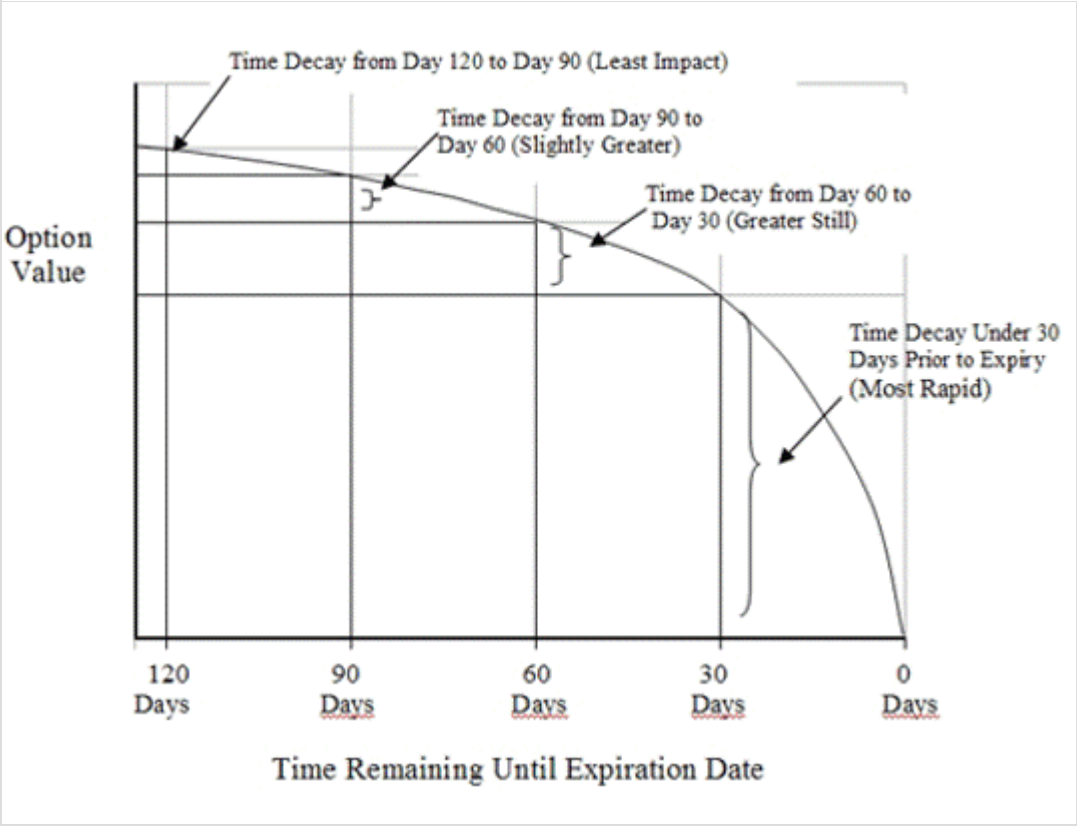
Gamma

Gamma is a valuable tool that helps determine the stability of Delta which can in turn forecast the probability of the option expiring in the money. Gamma is the rate of change in an option's delta per 1 point move in the underlying asset's price. If you are long an option, you are always long Gamma. If you are short an option, you are always short Gamma. Think of gamma as an acceleration factor, for example. When an option is near or ATM then Gamma is at its largest. When an option is deep in or deep out of the money then Gamma is at its lowest. When you buy an option, you want one with a large Gamma when entering a trade. So basically, when the option expires, you want the option to be deep in the money and have a low Gamma.

Theta

Theta also known as the "Time Decay" of an option is the dollar amount that an option will lose each day due to the passage of time. There is no way to mitigate this risk. Time decay is the greatest when the option is closest to expiration an ATM. Most options will expire worthless dues to Theta and

because many retail traders initiate for out the money options. This Greek is the most obvious but the most misunderstood. Theta is Inversely related to time.



It is very important to look at your Theta risk. Your trade idea must be right with in a specified time period and options often decay to zero right before your trade idea develops. **In other words, your view on an underlying's direction not only has to be right, but your timing must be right as well.**

Volatility & the Greeks

When **implied volatility goes up** in an option **Delta** does…

- Implied volatility goes up – OTM option delta goes up too – because since there is more volatility the likely hood of a big has more potential.
- Implied volatility goes up – ATM option delta does not change – it remains at or around 0.50 delta because there is still a 50/50 chance of it going up or down

When **implied volatility goes down** in an option **Delta** does…

- Implied volatility goes down – OTM option delta goes down or remains the same
- Implied volatility goes down – ATM option delta does not change – it remains at or around 0.50 delta because there is still a 50/50 chance of it going up or down

When **implied volatility goes up** in an option **Gamma** does…

- Implied volatility goes up – OTM option Gamma goes up too – because since there is more volatility likely hood of a big has more potential.
- Implied volatility goes up – ATM option Gamma goes down – because gamma is greatest when it is ATM.

When there is a greater chance for a larger move, Implied volatility goes up, and the Delta of OTM options goes up. Therefore, if the delta of OTM options goes up, the change in delta (Gamma) goes down because the lower the volatility the more potential there is in Gamma.

There is a linear relationship between days to maturity and theta. When volatility goes up, all options have more Theta. Theta is also greatest the closer you get to Maturity (Expiration date).

If you understand the Greeks, as a retail trader, you can make more informed decisions about which options to trade, when to trade them and profit from them.